Maternal FitBand: Project Gaps and Fast Path to Demo

Prepared as a prototype review. Estimates require clinical validation before any real-world use.

Overview

The overall design is sound, and most of the pieces exist. The gaps are in the hardware-to-model data path, the guideline-based alerting, and the mother-facing guidance.

What's missing

1. The hardware doesn't send what the models need (biggest gap)

The BP and glucose models take PPG waveform features: systolic and diastolic peak, pulse area, pulse amplitude, and pulse intervals. The firmware only sends heart rate, SpO2, and accelerometer values, so the server fills those inputs with defaults. BP and glucose are therefore not really computed from the sensor right now.

Fix: Change the firmware to stream the raw IR and red PPG samples over USB at about 100 Hz, with timestamps. Compute the PPG features in Python on the laptop, using the same definitions as the training data.

2. Activity and sleep need real windowed motion features

The activity and sleep models currently receive fixed accelerometer values, not features from the actual motion.

Fix: Compute features over a sliding window of about 5 to 10 seconds from the accelerometer data: mean and standard deviation of each axis, and the vector magnitude. Use those as model inputs.

3. Training isn't reproducible

The shipped models work, but the training script has wrong paths and skips the glucose step silently. Nothing documents the exact feature order each model expects.

Fix (fast): Don't retrain. Write a `feature_schema.json` that lists each model's inputs in order, and have the server check against it. This takes about an hour and prevents feature-mismatch bugs from returning.

4. Alerts aren't grounded in guidelines

Risk levels currently come from ad hoc thresholds, and mother-facing advice is free text generated by Gemini. That is not acceptable for medical guidance.

Fix: Build a rule-based guideline layer in Python. Gemini should only rephrase decisions already made, in a reassuring tone. It should never decide the alert level.

Example thresholds to check against current sources:

- **Blood pressure (ACOG):** 140/90 or higher is hypertension in pregnancy. 160/110 or higher is severe and needs urgent care.
- **Preeclampsia warning symptoms** (headache, vision changes, upper-right abdominal pain, shortness of breath, sudden swelling) should always trigger "contact your doctor or go to emergency care," whatever the sensor reads.
- **Glucose:** The app cannot diagnose gestational diabetes, because diagnosis requires a lab test. Out-of-range values should prompt "discuss this with your doctor," not a diagnosis.

Check every threshold against current ACOG, WHO, and NICE guidance before the demo.

5. The mother-facing screen

The dashboard should show each value in plain language: a color, a short sentence, and what to do next. It should also show a short list of emergency signs at the top, and a clear note that the device is a prototype and does not replace medical care.

6. Hardware connection (USB only)

Use the USB path only. The Wi-Fi and BLE code is separate, so leave it out of the demo firmware. The server's serial reader already supports USB, so the missing piece is a USB-only firmware that sends PPG samples and accelerometer data as JSON lines.

Fastest path to the demo

Tonight (must-do)

1. Write the USB-only firmware: raw PPG samples and accelerometer data as JSON, at 100 Hz.
2. Add PPG feature extraction and motion windowing in Python.
3. Add the guideline rule layer for BP, red-flag symptoms, and glucose messages, with a clear disclaimer.
4. Test with the real device: sitting, lying on the back, and walking. Confirm the readings change.

Tomorrow morning

5. Simplify the mother-facing screen: one card per value, one alert banner, and a short next-steps box.
6. Rehearse the demo twice, with the simulator as backup.

After the demo (don't attempt before it)

Clinical validation of BP and glucose from PPG, and retraining on a clean pipeline.

How to present this honestly

Frame the project as a prototype: a wearable that gathers maternal vitals and turns them into guideline-based guidance for mothers. State clearly that BP and glucose values are estimates needing clinical validation, and that the app does not replace a doctor. A demo that is accurate about its limits will be more convincing to an examiner than one claiming clinical accuracy it cannot yet support.